

СОЗДАНИЕ СЕТЕЙ

Расширьте возможности своей организации с помощью программно-конфигурируемой глобальной сети

Развертывание SD-WAN может значительно сократить счета за обслуживание, а то, что оно предлагает в виде гарантированного восстановления работоспособности, практически несуществующих простоев и абсолютной простоты установки, заставило бы любого руководителя C-suite насторожиться.

4 ноября 2019 года, 12:03 [Ашватт Нагарадж](#)



SD-WAN (программно-определяемая глобальная сеть) на первый взгляд кажется почти недостижимой мечтой о «более дешевом, быстром и качественном» способе соединения филиалов друг с другом и с центральным узлом или штаб-квартирой. Благодаря соединению нескольких филиалов с центром с помощью недорогих сервисов и централизованному управлению потоками данных на основе политик SD-WAN коренным образом изменит

преимущества с точки зрения оперативности, гибкости и простоты.

Развертывание SD-WAN может значительно сократить счета за обслуживание глобальной сети, а то, что оно предлагает в виде гарантированного восстановления работоспособности, практически несуществующих простоев и абсолютной простоты установки — за несколько дней, а не недель, — заставило бы любого руководителя C-suite насторожиться.

Сделаем шаг назад: что такое SD-WAN?

Сети SD-WAN объединяют все ваши филиалы, центры обработки данных и облачные сервисы в ячеистую систему, которая превращает все локации в единый пул, доступный для всех сервисов. Благодаря распределению вычислительной мощности и доступа по всей сети, а не по отдельным локациям, инфраструктура SD-WAN делает всю сеть более эффективной и быстрой. Наложение логических глобальных сетей позволяет организации сократить время ожидания и определить потребности в зависимости от местоположения.

Например, банк может привезти мобильный банкомат на музыкальный фестиваль или в район, пострадавший от стихийного бедствия, чтобы обеспечить посетителям или местным жителям удобный доступ к наличным, которого у них в противном случае не было бы. Этому «отделению» не потребуется такой же доступ к сервисам и программному обеспечению, как кассирам в главном офисе банка или сотрудникам отдела выдачи кредитов, поэтому программно-определяемая глобальная сеть сможет выделять ресурсы данных только для необходимых сервисов — выдачи наличных и доступа к главной бухгалтерской книге — и не уделять приоритетного внимания ненужным сервисам, таким как выдача кредитов.

Policies for each machine, whether a PC, a router, optimizer or any other networked device, are centralized. Behaviors for each can be defined and implemented across the entire network without requiring machine-by-machine configuration. And, with fully-managed SD-WAN, this can be done remotely by the service provider, delivering even greater savings in labor costs.

SD-WAN deployments fall into two categories — overlay and in-net. Overlay SD-WAN devices can (in some/most cases) replace entire **network stacks**, either in routers or replacing them as the branch WAN connection. Because this becomes a layer of enterprise infrastructure, it is easy to swap out network service

and control. In-net SD-WAN is also known as managed SD-WAN. In this case, a branch location would connect to the nearest Point of Presence (PoP) and the data is served up by the provider's infrastructure.

Why Implement SD-WAN?

Many WAN-enabled organizations use MPLS, but an April 2019 study by IDC found that 78% of organizations have either deployed SD-WAN technology or plan to do so in the next 12 months. This does not necessarily mean MPLS is going the way of the dodo. In fact, it may not be strategically prudent to simply replace an organization's MPLS infrastructure wholesale. It may be best to gradually substitute in SD-WAN infrastructure over a period of time, as lower-cost connectivity options present themselves, including business internet services or even consumer-grade broadband. One major area of consideration is an organization's future growth needs. Those looking for greater savings over a long timeframe would be wise to look into SD-WAN, for example planning all-new branches to be integrated with SD-WAN, and then upgrading older branches when necessary.

Because there is not a large amount of infrastructure involved — In many cases, one simple box is all that's needed, not a full stack of machines — deploying SD-WAN is much faster than legacy WAN solutions. A new branch could be up and running in 24 hours to a week, versus 30-90 days of installation and configuration with traditional approaches. It's easy to see how every day's delay in go-live time translates to lost revenue.

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transformation — being able to easily upgrade and incorporate new technologies as they come available. Data flows from branches can easily be crunched and analyzed at the data center or HQ, and real-time information can flow back. For example, a clothing retailer can offer free WiFi in their stores, and when a loyal customer enters and connects, the SD-WAN system can send the mobile device information back to the cloud, where it is correlated with that person's account and instant discounts or special offers can be sent to that device for in-store use. If an item is not available at that location, the inventory system can be alerted and shipment arranged, all to make the customer's experience that much greater. In a retail situation such as this, managed SD-WAN can prioritize application bandwidth on a branch-by-branch basis, so a west-coast location may require more mobile-friendly applications while shoppers are still in the store, and the east-coast branches shift resource allocation to the inventory database as they count items after store closing.

What's in It for Me?

Because of the flexibility of infrastructure required in overlay or in-net SD-WAN, major cost savings are common. In some cases, simply running the locally available commercial internet through an SD-WAN "box" is all that's needed. In others, a combination of an SLA-managed SD-WAN core with broadband internet access will be more appropriate. The beauty of SD-WAN is that it can use any available network routes, even legacy MPLS, to deliver the promised level of service. Where large, inexpensive internet links are prevalent, traffic can shift there, de-prioritizing expensive MPLS links. This range of connectivity options means cost curves can shift downward over time. The much smaller stack footprint — moving from a server room to a single box — brings capital costs down, as well.

IT departments always seem stretched thin on resources, even when fully staffed. Switching to a fully managed SD-WAN deployment with 24/7 support anywhere in the world has helped some organizations realize a 90 percent time savings on **network troubleshooting**.

In the end, it's up to every organization to put serious thought into whether SD-WAN is right for them. But given the ease of installation, the savings and the ability to not only be up and running in record time, but able to give your customers the experience they want today, it would behoove any IT department to look into it.

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